

From nitrogen metrics to decisions: building the implementation architecture for action

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KEY CLAIM

Nitrogen metrics become actionable not simply by being better integrated with each other, but when the right information reaches the right actor at a real decision point through trusted decision-support, advisory and governance arrangements.

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Nitrogen footprints, nitrogen use efficiency (NUE), and national or regional nitrogen budgets provide different views of the nitrogen system. Footprints connect nitrogen losses with consumption and supply chains (Leach et al., 2012); NUE assesses how efficiently nitrogen inputs are converted into useful outputs (EU Nitrogen Expert Panel, 2015); and nitrogen budgets reveal flows, stocks, losses and potential intervention points (Winiwarter et al., 2025). These functions are complementary, but they do not automatically translate into action. The practical starting point should therefore not always be “How can we integrate the metrics?”, but “What decision needs to change, who can change it, and what information is required?” Nitrogen governance spans actors, sectors and scales, while farmer decisions are shaped by regulation, markets, advisory systems and other forces beyond the farm (Kanter et al., 2025). At farm level, action ultimately takes the form of choices about fields, crops, rotations, nutrient management and mitigation under agronomic, economic and regulatory constraints.

Decision-support tools (DSTs) can provide a translation layer between nitrogen knowledge and such choices. Research on agricultural DSTs shows that technical performance alone is insufficient: tools must fit real decision contexts, accommodate user knowledge and support interpretation and action (Jakku and Thorburn, 2010; Rose et al., 2016). Their function is therefore not merely to calculate an optimum, but to make a decision space visible. Our work in TRANS4NUM illustrates this through spatial scenario tools that combine environmental, production and economic information with user-defined goals and constraints. PlantPerform now takes this logic into Danish advisory practice under new spatially differentiated nitrogen regulation. Here the chain is policy and environmental targets → farm and field data → modelled scenarios → adviser–farmer dialogue → field-level choices → monitoring and learning. The DST does not make the decision; it creates a transparent space in which modelled effects, regulation, economics, adviser expertise and farmer knowledge can be brought together. This suggests that integration should be functional rather than necessarily mathematical. Different nitrogen metrics can retain their distinct purposes while selected information moves between scales according to the decision being supported. The challenge is therefore to build an implementation architecture around the metrics: transparent models, trusted intermediaries, feasible incentives, clear institutional responsibilities and feedback from implementation into models, metrics and policy.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

The session contributions represent complementary elements of such an implementation architecture. National nitrogen budgets identify system-level flows and intervention points; improved interpretation of NUE helps distinguish management performance from structural changes in production; and production-informed footprints can retain production-side differences and connect them with downstream demand. The Danish footprint contribution further shows why decision support should not optimise one indicator in isolation: improved efficiency can coexist with high absolute nitrogen pressure, expanding production and cross-border nitrogen dependencies. My-N-Print demonstrates a parallel translation challenge at the consumer end of the system. Together, these perspectives suggest that the central integration question is not only how metrics relate to each other, but how the relevant information reaches actors in a form that can alter real decisions.

QUESTIONS TO SEED THE DISCUSSION

1. **Should integration start from the metrics, or from the decisions we want to influence: who needs to act, what can they change, and which nitrogen information do they actually need?**
2. **How can implementation outcomes feed back into metrics, models and policy so that nitrogen decision support becomes an adaptive learning system rather than a one-way transfer of information?**

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